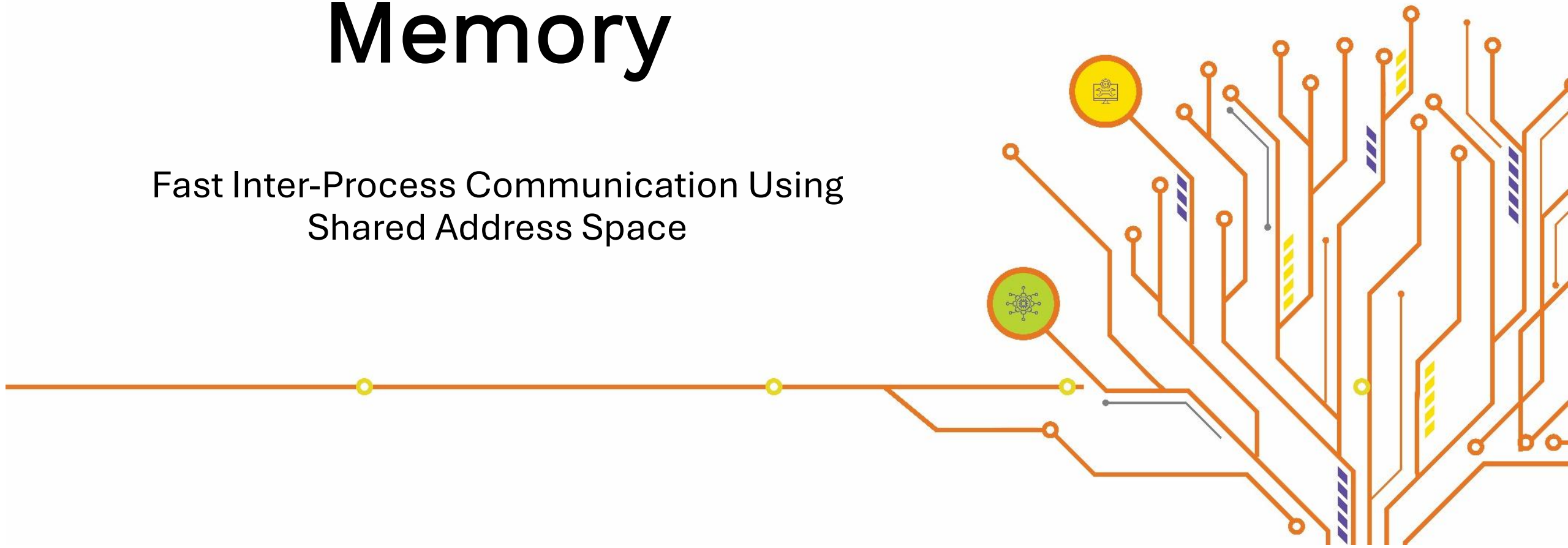
KPIT | APEX LAB

POSIX Shared Memory

Fast Inter-Process Communication Using Shared Address Space



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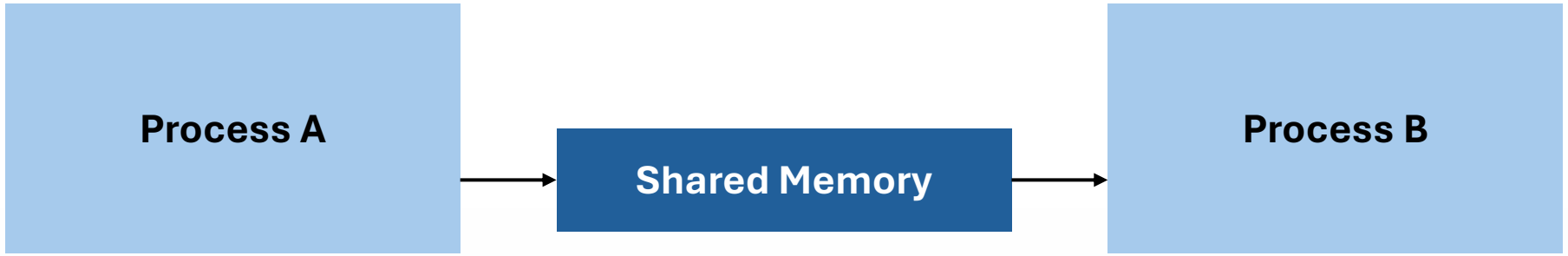
What is Shared Memory?

- Fastest IPC mechanism
- Multiple processes **map the same memory region**
- Data is exchanged by **direct read/write**, not system calls
- Created using **shm_open() + mmap()**
- Shared region lives inside **kernel-managed memory objects**



Why Use Shared Memory?

Highest throughput IPC

Zero-copy → no kernel copying overhead

Ideal for large data transfers

Great for:

- sensor buffers
- image frames
- circular queues
- control blocks
- shared lookup tables

POSIX Shared Memory Basics

- Identified by a **name** beginning with /

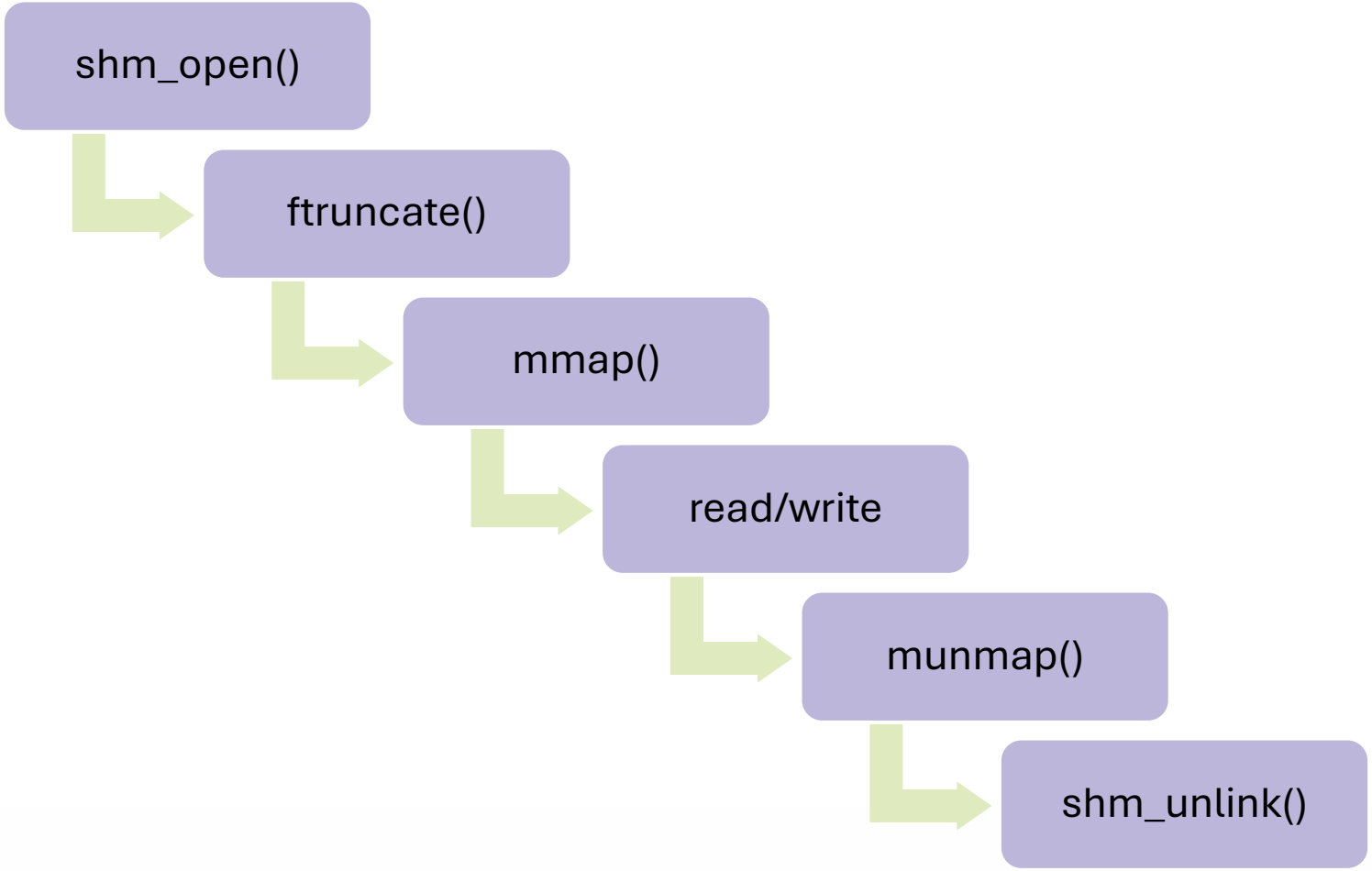
  Example: /apex_shm

- Behaves like a **file descriptor**
- Must specify size using ftruncate()
- Mapped into virtual memory using mmap()
- Must be cleaned using shm_unlink()

Key APIs

- **Create/Open:**
  - int fd = shm_open("/apex", O_CREAT | O_RDWR, 0666);
- **Set Size:**
  - ftruncate(fd, SIZE);
- **Map Memory:**
  - void *ptr = mmap(NULL, SIZE, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);
- **Cleanup:**
  - munmap(ptr, SIZE);
  - close(fd);
  - shm_unlink("/apex");.

Lifecycle of Shared Memory Object



Mapping Shared Memory

- mmap() parameters:
  - PROT_READ, PROT_WRITE
  - MAP_SHARED (changes visible to all processes)
- Important:
  - Use MAP_SHARED (not MAP_PRIVATE) for IPC
  - Mapping returns pointer → use it like a normal C array/struct

Example: Writer and Reader Process

```
int fd = shm_open("/apex", O_CREAT | O_RDWR, 0666);
ftruncate(fd, 1024);

char *p = mmap(NULL, 1024, PROT_READ|PROT_WRITE,
MAP_SHARED, fd, 0);
sprintf(p, "Hello from writer!");
```

```
int fd = shm_open("/apex", O_RDONLY, 0666);
char *p = mmap(NULL, 1024, PROT_READ, MAP_SHARED,
fd, 0);

printf("Reader got: %s\n", p);
```

Synchronization Required

Shared memory gives fast sharing but introduces race conditions.

- Examples requiring mutex/sem:
  - Updating shared counters
  - Writing multi-field structures
  - Producer-consumer buffers
  - Logging queues

Shared memory ≠ thread-safe.

Using Shared Memory with Mutex

Steps:
1. Put mutex structure **inside** shared memory
2. Initialize with `PTHREAD_PROCESS_SHARED` attribute
3. Multiple processes can lock/unlock it

Diagram:
[Shared Memory] → struct { mutex; data; }

Unmapping & Cleaning Up

- munmap(ptr, size) → detach mapping
- close(fd) → release descriptor
- shm_unlink(name) → delete shared-memory object
- Important:
  - unlink should be done by creator or cleanup tool.

Benefits vs Other IPC

IPC	Speed	Direction	Payload	Best Use
Pipes	Medium	One-way	Byte stream	Parent-child text data
Message Queues	Medium	Two-way	Structured msgs	Events, priorities
Shared Memory	**Fastest**	Both ways	Any data	Large data, real-time

Example 1: Basic Shared Memory (Writer → Reader)

Task:

- Writer creates `/lab_shm`, writes a string
- Reader opens same shared memory and prints it
- Cleanup using `shm_unlink()`

Example 2: Shared Integer Counter

Task:

- Writer initializes a counter to 0
- Reader process increments every second
- Writer prints current value every second
- Observe shared updates

Learning: shared variables without copying

Example 3: Producer–Consumer Buffer

Shared memory contains:

- circular buffer (array)
- two indexes: head, tail
- two semaphores (empty, full)

Implement: Producer writes items; consumer prints them.

Learning: synchronization + shared data structure

Real-World Usa Cases

- Shared vision/image buffer between processes
- Sensor data accumulation shared with control loop
- Shared configuration block for real-time apps
- Logging service shared across modules
- Robotics (shared state, map, telemetry)

Common Pitfalls

- Forgetting ftruncate() → 0-size mapping
- Using MAP_PRIVATE → changes not shared
- Not synchronizing writes → corrupted structures
- Forgetting shm_unlink() → orphaned objects remain
- Writing beyond allocated size → segmentation faults
- Multiple writers without locks → race conditions

Troubleshooting

- Use ls -l /dev/shm to list shared memory objects
- Use ipcs -m (System V, optional for comparison)
- Check return values of all system calls
- Use strace for debugging open/mmap calls
- Use hexdump to inspect raw memory if needed

Summary

- Fastest inter-process communication method
- shm_open + mmap create shared memory region
- Multiple processes map same address space → read/write directly
- Requires explicit synchronization
- Ideal for high-performance embedded systems

Scenario: Telemetry Ring Buffer with Dual Semaphores (1/4)

Context

- Implement a high-rate telemetry pipeline where a producer process samples data (e.g., random sensor values) and writes into a shared memory circular buffer; a consumer process reads, computes rolling statistics (min/max/avg), and prints them.
- Synchronization via counting semaphores empty and full (+ a mutex for head/tail if you choose).

Scenario: Telemetry Ring Buffer with Dual Semaphores (2/4)

PART 1 — Shared Region & Metadata

- Create telemetry_common.h, producer.c, consumer.c.
- Shared memory object /apex_shm_telemetry, size enough for:
  - struct header { int head, tail; int capacity; }
  - struct sample { uint64_t ts_ns; float value; } buf[CAP];
- Initialize with ftruncate and mmap in producer (creator).

Scenario: Telemetry Ring Buffer with Dual Semaphores (3/4)

PART 2 — Synchronization

- Semaphores in shared memory or named semaphores:
    - empty = CAP, full = 0.
- Producer flow (loop @ 100 Hz):
    - sem_wait(empty) → write sample at head → advance head (mod CAP) → sem_post(full)
- Consumer flow (loop):
    - sem_wait(full) → read sample at tail → advance tail → sem_post(empty)
    - Update sliding window stats and print every second.

Scenario: Telemetry Ring Buffer with Dual Semaphores (4/4)

PART 3 — Graceful Shutdown

- Handle SIGINT: producer writes a shutdown flag in header; consumer exits when queue drained.
- Cleanup: munmap, close, shm_unlink (producer), sem_close/sem_unlink if named.